Formant Frequencies

Table 1. Mean data on fundamental frequency and the first three formant frequencies for vowels of American English produced by adult male talkers. The data are from (1) Peterson and Barney (1952), (2) Hillenbrand et al. (1995); (3) Zahorian and Jagharghi (1993); (4) Hagiwara (1995), (5) Yang (1996), and (6) Childers and Wu (1991).

		Vowel											
		i *heed*	I *hid*	e *hate*	E *head*	ae *had*	a *hot*	ɔ *hawed*	o *hoed*	U *hood*	u *who'd*	ʌ *hud*	ɝ *heard*
f_0	1	136	135	---	130	127	124	129	---	137	141	130	133
	2	138	13?	129	127	123	123	121	129	133	143	133	130
	3	---	---	---	---	---	---	---	---	---	---	---	---
	4	---	---	---	---	---	---	---	---	---	---	---	---
	5	136	130	128	132	126	125	128	129	135	135	127	130
	6	132	130	---	124	123	120	120	---	126	130	120	122
F1	1	270	390	---	530	660	730	570	---	440	300	640	490
	2	342	427	476	580	588	768	652	497	469	378	623	474
	3	272	410	---	550	656	749	637	456	439	324	---	445
	4	291	418	403	529	685	---		437	441	323	574	429
	5	286	409	469	531	687	638	663	498	446	333	592	490
	6	303	439	---	542	645	673	615	---	487	342	591	477
F2	1	2290	1990	----	1840	1720	1090	840	----	1020	870	1190	1350
	2	2322	2034	2089	1799	1952	1333	997	910	1122	997	1200	1379
	3	2209	1859	----	1740	1748	1192	1004	1176	1234	1396	----	1286
	4	2338	1808	2059	1670	1600	---	1248	1188	1366	1417	1415	1362
	5	2317	2012	2082	1900	1743	1051	1026	1127	1331	1393	1331	1363
	6	2172	1837	---	1690	1622	1098	990	---	1168	1067	1194	1276
F3	1	3010	2550	----	2480	2410	2440	2410	----	2240	2240	2390	1690
	2	3000	2684	2691	2605	2601	2522	2538	2459	2434	2343	2550	1710
	3	2971	2600	----	2535	2345	2501	2400	2307	2349	2352	----	1656
	4	2920	2588	2690	2528	2524	---	2441	2430	2446	2399	2496	1683
	5	3033	2671	2636	2561	2497	2318	2527	2375	2380	2282	2494	1787
	6	2851	2482	---	2456	2357	2457	2465	---	2307	2219	2401	1707

Table 2. Mean data on fundamental frequency and the first three formant frequencies for vowels of American English produced by adult female talkers. The data are from (1) Peterson and Barney (1952), (2) Hillenbrand et al. (1995); (3) Zahorian and Jagharghi (1993); (4) Hagiwara (1995); (5) Yang (1996), and Childers and Wu (1991).

		Vowel											
		i	I	e	E	ae	a	ɔ	o	U	u	ʌ	ɝ
		heed	*hid*	*hate*	*head*	*hat*	*hot*	*hawed*	*hoed*	*hood*	*who'd*	*hud*	*heard*
f_0	1	235	232	---	223	210	212	216	---	232	231	221	218
	2	227	224	219	214	215	215	210	217	230	235	218	217
	3	---	---	---	---	---	---	---	---	---	---	---	---
	4	---	---	---	---	---	---	---	---	---	---	---	---
	5	221	216	209	211	209	205	206	207	214	228	206	211
	6	233	228	---	219	216	214	216	---	220	222	215	217
F1	1	310	430	---	610	860	850	590	---	470	370	760	500
	2	437	483	536	731	669	936	781	555	519	459	753	523
	3	338	486	---	745	922	981	793	532	528	400	---	542
	4	362	467	440	806	1017	---	947	516	486	395	847	477
	5	390	466	521	631	825	782	777	528	491	417	701	523
	6	378	512	---	661	842	838	745	---	522	409	724	558
F2	1	2790	2480	----	2330	2050	1220	920	----	1160	950	1400	1640
	2	2761	2365	2530	2058	2349	1551	1136	1035	1225	1105	1426	1588
	3	2837	2284	----	2123	2089	1440	1176	1419	1437	1617	----	1532
	4	2897	2400	2655	2152	1810	---	1390	1392	1665	1700	1753	1558
	5	2826	2373	2536	2244	2059	1287	1140	1206	1486	1511	1641	1550
	6	2586	2197	---	2013	1933	1246	1190	---	1386	1361	1445	1504
F3	1	3310	3070	----	2990	2850	2810	2710	----	2680	2670	2780	1960
	2	3372	3053	3047	2979	2972	2815	2824	2828	2827	2735	2933	1929
	3	3456	3093	----	3041	2981	2847	2860	2789	2848	2766	----	1992
	4	3495	3187	3252	3064	2826	---	2725	2903	2926	2866	2988	1995
	5	3416	3014	2991	2968	2928	2563	2895	2836	2836	2796	2901	1927
	6	3286	2996	---	2956	2982	2945	2853	---	2792	2730	2863	2024

Fundamental Frequency

Table 6–5A. Speaking Fundamental Frequency of Normal Children

Age (y)		*No. of*	*Mean F_o**			*SD*	
Mean	*Range*	*Subj.*	*(Hz)*	*(ST)*	*Range*	*(ST)*	*Sources*
			Boys				
Spontaneous Speech							
3.8	3.2–4.6	10	283	49.4	241–322		10
4.6		15	252.4	47.4	217.2–292.7		4
‡5.4	5.1–6.0	16	249.0		218.2–287.5	5.39	8
††5.5	5.0–6.0	18	241.3		204.9–274.3	5.26	8
5.6	5.1–6.3	15	240.1	46.5	211.9–263.1	4.38	8
††	6.0–6.9	50	219.5	44.9	134.8–298.7		9
6.5		14	247.3	47.0	204.1–274.4		4
††8.3		15	230	45.8		2.1	11
8.4		15	213	44.4		2.5	11
††9.5		15	217	44.8		2.5	11
9.4		15	219	44.9		1.9	11
††10.6		15	204	43.7		3.2	11
10.5		15	220	45.0		2.3	11
Reading							
7.0	6.8–7.2	15	294	50.0		2.2	2
8.0	7.8–8.1	15	297	50.2		2.0	2
††8.3		15	233	46.0		2.1	11
8.4		15	220	45.0		2.1	11
††9.5		15	232	45.9		2.6	11
9.4		15	232	45.9		1.9	11
††10.6		15	215	44.6		3.4	11
10.5		15	225	45.4		2.5	11
10.0	9.8–10.2	6	269.7	48.7		2.38	3
	8.7–12.9	19	273	48.7			7
11.2	10–12	18	226.5	45.5	192.1–268.5	1.51	5
14.2	13.9–14.3	6	241.5	46.8		3.4	3
	13–15.9	15	184	41.9			7
	16–19.5	14	125	35.2			7
			Girls				
Spontaneous Speech							
5.5		18	247.6	47.0	211.9–295.2		4
5.6	5.1–6.1	20	243.4	46.7	195.2–291.1	5.59	8
††5.6	5.1–6.3	17	231.5	45.9	208.1–261.7	5.03	8
‡5.6	5.1–5.9	19	248.0	47.1	217.6–274.0	4.64	8
††	6.0–6.9	50	211.3		137.6–297.5		9
6.4		19	247.0	47.0	217.7–274.1		4
Reading							
7.0	6.8–7.2	15	281	49.2		2.0	1
7.9	7.8–8.1	15	288	49.6		2.8	1
	8.6–12.9		256	47.6			6
11.2	10–12	18	237.5	46.32	198.1–271.1	1.51	5
	13–15.9		248	47.1			6
	16—19.8		241	46.6			6

(Continued)

Fundamental Frequency (cont.)

Table 6–4. Mean Phonational F_o of Normal Aged Speakers

Age Range	*Mean Age*	*No. of Subj.*	*Mean SF_o (Hz)*	*SD (Hz)*	*Sustained /ɑ/ mean F_o (Hz)*	*SD or range of sustained /ɑ/ (Hz)*	*Source*
			Men				
60–69		21			118	15.8	2
69–85	75	20	162	30.7	162	32.4	1
60–92	ca. 73	6	126.2	25.8			8
> 65		18	130.1	16.7			9
65–85	75	15	127	3.1			11
70–79		27			121	26.9	2
80–92		22			127	22.7	2
	75.3	10			132*	20.2	5
	88	7			126	110–142	3
			Women				
60–69		36			179	27.7	2
60–77	69	20			191.8†	155–215 (SD = 13.9)	7
60–92	ca. 73	15	172.65	13.6			8
69–85	75	20	177	26.7	165	32.5	1
65–85	79	19	175	2.4			11
70–79		54			186	29.4	2
	74.6	11			211*	31.5	5
75–90	79.4	17	175.3	13.4	176.7	19.04	6
80–96		67			178	30.7	2
	88	18			181	131–228	3
100–107	102	5			140†	19.5†	4
	105	1	219	78			10

* Vowel /æ/

† Vowel /i/

Sources:

1. Honjo and Isshiki, 1980. F_o by spectrum analysis. F_o data gathered with perceptual ratings of voice quality.
2. Mueller, Sweeney, and Baribeau, 1984. Subjects "free of major medical problems, particularly those affecting the pulmonary-laryngeal system, as well as communication disorder of significant hearing loss." F_o readout by Visi-Pitch® 6087A system.
3. Mueller, 1982. "Only persons who were free of major medical problems, particularly those affecting the larynx or tracheobronchial system were included. Subjects who exhibited speech, voice, language and/or significant hearing difficulties were excluded." Maximally sustained /a/ in "normal speaking voice." Maximum phonation time also assessed.
4. Mueller, 1989. Subjects were "able to hear speech at normal conversational levels, were free of speech or language defects, were lucid and oriented to the task and did not suffer from dementia. . . . None . . . were afflicted with neurological disease and/or distress of the upper respiratory tract." 9-second sample of extemporaneous speech, evaluated by Visi-Pitch® system.
5. Higgins, M. B., and Saxman, J. H. A comparison of selected phonatory behaviors of healthy aged and young adults. *Journal of Speech and Hearing Research*, *34* (1991) 1000–1010.
6. Brown, Morris, and Michel, 1989. Subjects had no history of neurological or respiratory disease and had no hearing loss > 35 dB at 500, 1K, and 2K Hz in at least one ear. Sustained /a/ and second sentence of Rainbow Passage at conversational loudness. Analysis from tape recordings. Compared to young women, see Table 6–2.
7. Biever and Bless, 1989. Subjects were free of neurological disease, no history of laryngeal surgery or disorder, nonsmokers, vocally untrained, hearing threshold better than 35 dB at octave frequencies 500 to 4K Hz in better ear, between 62 and 68 inches (139 cm and 152 cm). Analysis of tape recorded samples.
8. Weatherley, Worrall, and Hickson, 1997. SF_0 calculated from spoken phrases consisting exclusively of voiced phones. Compared to hearing-impaired aged subjects.
9. Morris, Brown, Hicks, and Howell, 1995. Combined reading (Rainbow Passage) and spontaneous speech (picture description). Compared to younger men and to professional singers.
10. Max and Mueller, 1996. Conversational speech (radio interview) of an independently-living woman.
11. Brown, Morris, Hollien, and Howell, 1991. Reading of first paragraph of Rainbow Passage. Comparison to professional singers.

SPL

Table 5–10. Sound Pressure Levels (SPL) of Sustained Vowels at Different Frequency and Intensity Levels in the Modal Register

Sex	*F_o Level*	*Vowel*	*Intensity Level*	*dB SPL Mean (SD)*	*Source*
M	25%	/a/	Minimal	66	*a*
			Comfortable	81	*a*
			Maximal	95	*a*
M	50%	/a/	Minimal	70	*a*
			Comfortable	84	*a*
			Maximal	101	*a*
M	75%	/a/	Minimal	72	*a*
			Comfortable	86	*a*
			Maximal	98	*a*
F	Comfortable	/i/	"relatively soft"	75.4 (4.4)	*b*
F	Comfortable	/ɑ/	"relatively soft"	75.0 (4.5)	*b*
F	Comfortable	/u/	"relatively soft"	75.9 (4.5)	*b*

a. "Vocal Intensity in the Modal and Loft Registers," by R. H. Colton, 1973, *Folia Phoniatrica*, *25*, 62–70. 8 "young adult" males, vocally untrained. Frequency levels are percent of total phonational frequency range in the modal register. 9 inch (≈ 23 cm) microphone-to-mouth distance. Compared to falsetto register and to trained singers.

b. "Effects of Prolonged Loud Reading on Selected Measures of Vocal Function in Trained and Untrained Singers," by M. P. Gelfer, M. L. Andrews, and C. P. Schmidt, 1991, *Journal of Voice*, *5*, 158–167. 24 "Young adult" women without vocal training. 5 inch (≈ 13 cm) microphone-to-mouth distance. Compared to performance after 1 hour of reading and to trained singers.

tation system to display vocal intensity as a function of vocal F_0 on an oscilloscope screen has been designed by Komiyama, Watanabe, Nishinow, and Yanai, 1977.)

The interrelationship of glottal adjustment and driving pressure underlies a graph showing a speaker's minimum and maximum measured phonatory intensity at frequencies that cover his F_0 range. Data plots of this type, introduced by Stout (1938) and explored early on by Calvet and Malhiac (1952) and Damsté (1970), have become very much more popular in recent years. An example is shown in Figure 5–8. A plot of this type has been referred to by several different names—including *fundamental frequency-sound pressure level profile* (Coleman, Mabis, & Hinson, 1977), *Stimmfeld* (Rauhut, Stürzbecher, Wagner, & Seidner, 1979; Klingholz & Martin, 1983; Klingholz, Martin, & Jolk, 1985; Seidner, Krüger, & Wernecke, 1985), *voice profile* (Bloothooft, 1982), *phonogram* (Komiyama, Watanabe, & Ryu, 1984), and *phonetogram* (Damsté, 1970; Schutte & Seidner, 1983; Gramming & Sundberg, 1988). Those terms, for various reasons, have been felt to be less than optimal. In a report published in 1992, the Committee on Voice of the International Association of Logopedics and Phoniatrics recommended that they be replaced by the name ***voice range profile*** (VRP)—or its direct translation in other languages (Bless & Baken, 1992). That recommendation has been accepted for the discussion that follows.[11]

VRP Test Procedure

Despite its increasing popularity, there is effectively no standardization of either the method by which the data for the VRP are to be obtained or the manner in which

[11] An overall review of the voice range profile and various factors affecting is has been published by Gramming (1991).

Jitter & shimmer for selected vowels

TABLE 1. Mean jitter, shimmer, and signal-to-noise (SNR) values obtained for sustained phonations associated with different frequencies (speaking fundamental frequency, or SFF, and one octave above SFF), intensities (60, 70, and 80 dB), and vowels (/i/ and /ɑ/). The standard deviation associated with each measure is in parentheses. These data are based on the vocal productions of 29 female subjects.

	Jitter	Shimmer	SNR
SFF			
60 dB			
/i/	.688 (.395)	2.91 (1.32)	19.9 (3.5)
/ɑ/	.714 (.481)	3.92 (2.13)	16.3 (3.7)
70 dB			
/i/	.305 (.116)	1.46 (0.57)	21.3 (3.0)
/ɑ/	.333 (.105)	2.53 (0.86)	22.1 (2.9)
80 dB			
/i/	.217 (.053)	1.86 (0.71)	20.3 (2.1)
/ɑ/	.259 (.088)	1.70 (0.60)	25.4 (2.4)
One octave above SFF			
60 dB			
/i/	.433 (.182)	1.95 (0.77)	25.1 (4.1)
/ɑ/	.424 (.210)	2.37 (0.85)	23.0 (4.3)
70 dB			
/i/	.328 (.172)	1.19 (0.58)	25.8 (5.0)
/ɑ/	.317 (.139)	2.07 (0.81)	24.7 (3.8)
80 dB			
/i/	.261 (.141)	0.93 (0.32)	25.8 (3.9)
/ɑ/	.279 (.102)	1.90 (0.76)	25.9 (3.7)

Jitter Results

In order to determine the effects of frequency, intensity, and vowel selection on the phonatory stability measure of jitter, a univariate repeated measures ANOVA was performed. The results are displayed in Table 2. Main effects for both frequency and intensity were significant at the .01 level or better, as was a frequency-by-intensity interaction. No other main effects or interactions were significant.

The significant frequency-by-intensity interaction, pictured in Figure 1, was followed up by post hoc testing using the Tukey procedure, so that significant differences between various frequency-intensity combinations could be assessed. The Tukey procedure was selected because it is not unnecessarily conservative (Hays, 1988, pp. 418–420) and because it permits comparisons between pairs of means (comparison among groups of means was not considered necessary). Results of post hoc testing revealed that low frequency-low intensity vocalizations had significantly higher levels of jitter (indicating greater cycle-to-cycle frequency perturbation) than any other frequency-intensity combination ($r = 6$, $df = 336$, critical difference = 0.194 at $p < .01$).

Shimmer Results

In order to determine the effects of frequency, intensity, and vowel selection on the phonatory stability measure of shimmer, a second univariate repeated measures ANOVA was performed. The results are displayed in Table 3. Main effects for frequency, intensity, and vowel were all significant at the .01 level or better, as were a frequency-by-intensity interaction, and a frequency-by-intensity-by-vowel interaction.

The highest-order significant interaction resulting from this analysis, frequency by intensity by vowel (pictured in Figure 2), was followed up by post hoc testing, again using the Tukey procedure (Hays, 1988), so that significant differences between various frequency-intensity-vowel combina-

TABLE 1. Means and standard deviations (in parenthesis) of mean f_0 in Hz, f_0 standard deviation in semitones, jitter in msec, jitter in percent, and shimmer in dB for each of eight vowels produced by 20 adult males.

Variables	i	ɪ	æ	ɑ	o (*Vowels*)	ʊ	u	ə	*Mean*
Mean f_0 (Hz)	128.5 (18.5)	126.8 (18.2)	125.2 (17.6)	125.0 (17.6)	126.4 (19.1)	127.0 (18.2)	128.8 (17.7)	126.8 (18.8)	126.8
f_0 standard deviation (semitone)	.18 (.09)	.27 (.15)	.26 (.15)	.25 (.14)	.29 (.17)	.25 (.20)	.19 (.10)	.27 (.19)	.25
Jitter (msec)	.054 (.019)	.058 (.023)	.059 (.017)	.054 (.019)	.074 (.045)	.069 (.038)	.053 (.028)	.071 (.050)	.062
Jitter (%)	.681% (.224)	.713% (.236)	.724% (.187)	.661% (.180)	.820% (.341)	.841% (.384)	.662% (.262)	.864% (.507)	.746%
Shimmer (dB)	.168 (.065)	.145 (.032)	.165 (.086)	.132 (.023)	.167 (.081)	.194 (.114)	.137 (.029)	.248 (.252)	.170

DDKs (AMRs)

TABLE 8. Normative adult data on MRR in syllables/second for [pʌ]. M = male; F = female; YA = young adult; A = adult with unreported or wide range of age; GA = geriatric adult.

Source	*Subjects*	*M*	*SD*	*Range*
Dworkin et al. (1980)	M, A	6.5	–	4.5–7.5
	F, A	6.1	–	4.6–8.6
Kreul (1972)	M, YA	6.0	0.66	–
	F, YA	5.9	0.35	–
	M, GA	6.0	0.82	–
	F, GA	6.7	0.93	–
Lass & Sandusky (1971)	M, YA	6.3	0.53	5.2–7.4
	F, YA	6.2	0.61	5.0–7.8
Lundeen (1950)	M & F, YA	7.0	–	–
Ptacek et al. (1966)	M, YA	7.0	1.0	5.4–9.4
	F, YA	6.9	0.6	5.8–7.8
	M, GA	5.4	1.2	2.8–8.2
	F, GA	5.0	1.2	1.3–7.0
Sigurd (1973)	8M & 1F, A	6.9	–	–
Tiffany (1980)	7M & 3F, A	7.1	0.7	–

TABLE 10. Normative adult data on MRR in syllables/second for [kʌ]. M = male; F = female; YA = young adult; A = adult with unreported or wide range of age; GA = geriatric adult.

Source	*Subjects*	*M*	*SD*	*Range*
Dworkin et al. (1980)	M, A	6.1	–	4.4–7.5
	F, A	5.7	–	4.3–7.9
Kreul (1972)	M, YA	5.4	(0.54)	–
	F, YA	5.2	(0.60)	–
	M, GA	5.8	(0.62)	–
	F, GA	5.9	(0.83)	–
Lass & Sandusky (1971)	M, YA	5.8	0.55	4.6–6.8
	F, YA	5.6	0.75	4.0–7.4
Lundeen (1950)	M & F, YA	6.2	–	–
Ptacek et al. (1966)	M, YA	6.2	0.8	5.0–8.2
	F, YA	6.2	0.8	4.6–8.2
	M, GA	4.9	1.0	2.6–6.8
	F, GA	4.4	1.1	2.2–6.4
Sigurd (1973)	8M & 1F, A	6.4	–	–
Tiffany (1980)	7M & 3F, A	6.2	0.8	–

by increases in the ratio of maximum velocity to movement amplitude. One interpretation of this relationship is that the kinematic changes that occur with increased rate of movements are brought about by increased stiffness. That is, slow rates of repetition are associated with decreased stiffness (reflected by a small ratio of maximum velocity to movement amplitude), and fast rates of syllable repetition are associated with increased stiffness (and a larger ratio of maximum velocity to movement amplitude). One potential these considerations may hold for the assessment of speech is a kinematic analysis, with velocity and amplitude as the primary variables, that can be interpreted with respect to the biomechanical characteristics of the articulator.

A further prospect for dynamic analysis is a geometric type of analysis in which phase plane trajectories are generated by continuously plotting the relationship between the position of an articulator and its velocity. Examples of this kind of analysis are shown in Kelso, Vatikiotis-Bateson, Saltzman, and Kay (1985). Although the analyses described by Kelso et al. are for reiterant speech based on the syllable /ba/, the same principles of analysis would apply to syllable repetitions used in diadochokinesis. Indeed, a similar graphical analysis, based on a plot of intensity (I) versus rate of change in intensity ($\Delta I/\Delta t$), was used by von Cramon and Ziegler (in press) to study syllable repetitions in dysarthric speakers. Elaboration of these analyses is beyond the scope of this paper but is one subject of a paper in preparation.

TABLE 9. Normative adult data on MRR in syllables/second for [tʌ]. M = male; F = female; YA = young adult; A = adult with unreported or wide range of age; GA = geriatric adult.

Source	*Subjects*	*M*	*SD*	*Range*
Dworkin et al. (1980)	M, A	6.5	–	4.4–8.2
	F, A	6.0	–	4.3–8.5
Kreul (1972)	M, YA	6.0	0.96	–
	F, YA	5.8	0.37	–
	M, GA	5.8	0.69	–
	F, GA	6.5	0.44	–
Lass & Sandusky (1971)	M, YA	6.1	0.50	5.0–7.8
	F, YA	6.2	0.64	4.4–7.8
Lundeen (1950)	M & F, YA	7.1	–	–
Ptacek et al. (1966)	M, YA	6.9	1.1	4.2–9.4
	F, YA	6.8	1.0	4.8–8.4
	M, GA	5.3	1.0	3.0–6.8
	F, GA	4.8	1.1	2.2–6.8
Sigurd (1973)	8M & 1F, A	6.9	–	–
Tiffany (1980)	7M & 3F, A	7.1	0.8	–

GENERAL CONCLUSIONS

Maximum performance tests find some of their most frequent clinical applications in screening and in the assessment of motor speech disorders. They have been used to identify diminished strength, speed, and range. Even though the requirements of speech may be well within the maximal performances of normal speakers, it can be important to determine when a disordered talker

TABLE 11. Normative data on MRR in syllables/second for [pʌtəkə]. M = male; F = female; YA = young adult; A = adult with unreported or wide range of age; GA = geriatric adult; B = boy; G = girl.

Source	*Subjects*	*M*	*SD*	*Range*
Lass & Sandusky (1971)	M, YA	6.36	0.42	4.8–7.2
	F, YA	6.27	0.31	4.8–7.2
Tiffany (1980)	7M & 3F, A	7.5	1.1	–
Ptacek et al. (1966)	M, YA	5.8	1.0	4.8–8.2
	F, YA	6.3	0.9	3.8–7.8
	M, GA	4.4	1.3	2.4–7.0
	F, GA	3.6	1.3	1.4–6.2
Blomquist (1950)	B-9	4.3	0.68	
	G-9	4.8	0.72	
	B-10	5.0	0.80	
	G-10	5.0	0.47	
	B-11	5.0	0.54	
	G-11	5.3	0.58	

Max. phonation duration

TABLE 2. Normative data on maximum phonation duration (in seconds except for the coefficient of variation, C, which is dimensionless) for vowel /a/. M = male; F = female.

Source	*Subjects*	*Sex*	*M*	*SD*	*Range*	*C*
Harden & Looney (1984)	6-year-olds	M	10.4	5.1	3.8 –16.8	.49
		F	10.6	6.3	6.2 –30.6	.59
Beckett et al. (1971)	7-year-olds	M	14.2	3.3	12.0 –22.0	.23
		F	15.4	2.7	9.0 –19.0	.175
Finnegan (1984)	3-year-olds	M	7.9	1.81	4.38–11.46	.23
	3-year-olds	F	6.3	1.76	2.84– 9.72	.28
	4-year-olds	M	10.0	2.51	5.08–14.90	.25
	4-year-olds	F	8.7	1.84	5.26–12.46	.21
	5-year-olds	M	10.1	3.05	4.15–16.09	.30
	5-year-olds	F	10.5	2.57	5.44–15.50	.24
	6-year-olds	M	13.9	2.98	8.06–19.74	.21
	6-year-olds	F	13.8	3.65	6.66–20.96	.26
	7-year-olds	M	14.6	2.82	9.11–20.15	.19
	7-year-olds	F	13.7	2.45	8.88–18.48	.18
	8-year-olds	M	16.8	4.51	7.98–25.64	.27
	8-year-olds	F	17.1	4.62	8.07–26.17	.27
	9-year-olds	M	16.8	6.07	4.94–28.72	.36
	9-year-olds	F	14.5	3.78	7.07–21.87	.26
	10-year-olds	M	22.2	4.74	12.91–31.49	.21
	10-year-olds	F	15.9	5.99	4.14–27.62	.38
	11-year-olds	M	19.8	3.79	12.43–27.27	.19
	11-year-olds	F	14.8	2.06	10.73–18.79	.14
	12-year-olds	M	20.2	5.72	9.02–31.44	.28
	12-year-olds	F	15.2	3.87	7.58–22.74	.25
	13-year-olds	M	22.3	8.19	6.29–38.29	.37
	13-year-olds	F	19.2	4.58	10.27–28.21	.24
	14-year-olds	M	22.3	6.89	8.84–35.84	.31
	14-year-olds	F	18.8	5.15	8.76–28.94	.27
	15-year-olds	M	20.7	5.32	10.32–31.16	.26
	15-year-olds	F	19.5	4.66	10.40–29.93	.24
	16-year-olds	M	21.0	4.40	12.43–29.66	.21
	16-year-olds	F	21.8	4.47	13.09–30.61	.20
	17-year-olds	M	28.7	7.08	14.83–42.57	.25
	17-year-olds	F	22.0	6.30	9.65–34.33	.29
Lewis et al. (1982)	8-year-olds	M	20.0	–	11.5 –24.5	
	8-year-olds	F	19.1	–	11.9 –23.0	
	10-year-olds	M	24.9	–	15.9 –39.0	
	10-year-olds	F	16.5	–	12.9 –21.8	

phonatory function. It is beyond the scope and purpose of this paper to consider them in detail, but several possible analyses are described in the literature. If a spectrograph is available, then spectrograms can be used to describe or measure various features of the acoustic pattern (Rontal, Rontal, & Rolnick, 1975; Yanagihara, 1967). Vocal tremor, noise, voice breaks, and other irregularities are features to be noted. Using a graphic level recorder, the noise level in the spectrum can be estimated to derive the spectral noise level (SNL), defined as the lowest peak marking of the graphic level recorder stylus in each 100-Hz section of the spectrum (Arnold & Emanuel, 1979; Emanuel, Lively, & McCoy, 1973; Emanuel & Sansone, 1969; Emanuel & Whitehead, 1979; Hanson & Emanuel, 1979). Perturbations in fundamental frequency (jitter) and amplitude (shimmer) also can be derived from the sustained

Max. phonation duration

TABLE 2. continued

Source	*Subjects*	*Sex*	*M*	*SD*	*Range*	*C*
Williams (1977)	8-year-olds	M	13.3	2.5	–	.19
	8-year-olds	F	13.6	5.2	–	.38
	11-year-olds	M	17.8	4.7	–	.26
	11-year-olds	F	15.8	4.1	–	.26
Child (1979)	10-year-olds	M	20.2	4.7	–	.23
	10-year-olds	F	15.1	4.3	–	.28
Reich et al. (1986)	8.5–10.4 years	F	14.3	4.69	–	.33
Ptacek & Sander (1963)	Young adults	M	22.6	8.1	9.3–43.3	.36
		F	15.2	5.0	6.2–28.4	.33
Ptacek et al. (1966)	Young adults	M	24.6	6.7	12.5–36.0	.27
		F	20.9	5.7	11.8–32.0	.27
Kreul (1972)	Young adults		18.2	4.3	–	.24
Hirano et al. (1968)	Adults	M	34.6	–	15.0–62.3 (CR)	
		F	25.7	–	14.3–40.4 (CR)	
Inglis (1977)	Young adults	M	24.8	8.4	–	.34
		F	22.8	4.1	–	.18
Taylor (1980)	Young adults	M	28.0	8.9	–	.32
		F	22.9	5.8	–	.25
Neiman & Edeson (1981)	Young adults	M	29.0	5.5	–	.19
		F	19.6	4.7	–	.24
Yanagihara & Koike (1967)	Young adults	M	30.2	9.7	20.4–50.7	.32
		F	22.5	6.1	16.4–32.90	.27
Bless & Hirano (1982b)	Young adults	M	33.6	11.4	16.7–58.4	.34
		F	26.5	11.3	11.6–60.5	.43
Canter (1965)	Men (35–75 years)		20.6[a]	–	14.8–42.4	–
Kreul (1972)	Aged M (65–75 years)		14.6	5.9	–	.40
	F (66–93 years)		14.6	5.8	–	.40
Ptacek et al. (1966)	Aged M (68–89 years)		18.1	6.6	10.0–37.2	.36
	F (66–93 years)		14.2	5.6	7.0–24.8	.39
Mueller (1971)	Aged M (51–65 years)		13.0	–	–	–
	F (49–72 years)		15.4	–	–	–
Mueller (1982)	Aged M (85–92 years)		13.0	–	7.0–12.0	–
	F (85–96 years)		10.0	–	6.0–18.0	–

Note. CR = critical region.
[a]Median.

phonation. Normative data are available in several papers (e.g., Hollien, Michel, & Doherty, 1973; Horii, 1979, 1985; Klingholz & Martin, 1985; Koike, 1973; Ramig & Ringel, 1983; Sorenson & Horii, 1984). For sustained vowel phonation in the modal register, jitter averages about 1% and shimmer averages about 0.5 dB. Kojima, Gould, Lambiase, and Isshiki (1980) used Fourier analysis to derive a signal-to-noise (S/N) ratio; and Yumoto, Gould, and Baer (1982) described a harmonics-to-noise (H/N) ratio for voice analysis. Finally, digital processing techniques can be used to derive several measures of voice function (Davis, 1976). See Johnson (1984) for a helpful review of voice analyses.

Another voice analysis tool is the phonetogram, which describes the fundamental frequency and intensity ranges of a voice and is therefore a portrayal of vocal maximum performance. This analysis—which also has been named a voice profile, voice field, voice area, and $f_{o/SPL}$ profile—is discussed further in conjunction with maximum vocal intensity (to follow). Sonninen, Hurme, Toivonen, and Vilkman (1985) described a *voice field* description of vowel phonation. The voice field is a plot